

Modulo Synthesis

Volume I: The Geometry of Spacetime

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“He who thinks the continuum is real has not counted the bits.”

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Part I

The Arena

The Failure of the Continuum

Modulo Overview

- *Why “smoothness” is an illusion at the Planck scale.*
- *The fundamental conflict: Discreteness vs. Lorentz Symmetry.*
- *The Lattice Paradox: Why a grid breaks relativity.*
- *The Solution: Stochastic Discreteness (Poisson Sprinkling).*

1.1 The Premise

Imagine a beach. From a mile away, the sand looks like a smooth, continuous golden sheet. You can draw lines on it, slide across it, and treat it as a perfect mathematical plane. But if you kneel down and look closely, the “continuity” vanishes. You see individual grains of silica, jagged and discrete. There is no “half-grain”; there is sand, or there is vacuum.

Modern physics is stuck in the “mile-away” view. General Relativity (GR) models spacetime as a smooth manifold—a continuum that can be zoomed into infinitely. You can define coordinates x^μ to infinite precision, e.g., $x = 1.0000\dots$ meters.

Quantum Mechanics (QM), however, tells a different story. It says that action is quantized ($\hbar$). It suggests that measuring a position with infinite precision requires infinite energy, creating a black hole that swallows the experiment. This implies a fundamental limit to resolution—a “pixel size” for reality, known as the Planck length ($\ell_P \approx 1.6 \times 10^{-35}$ meters).

If spacetime has a minimum length, it cannot be continuous. It must be discrete.

This book explores the consequences of taking that discreteness seriously. Rather than “approximating” spacetime as discrete, we consider the hypothesis that it **is** discrete, and ask whether the continuum might be a large-scale illusion.

1.2 The Conflict: Discreteness vs. Relativity

Here is the problem. If you try to make spacetime discrete in the naive way—by turning it into a grid or lattice—you break the most important symmetry in physics: **Lorentz Invariance**.

Definition 1: Lorentz Invariance

The laws of physics are the same for all observers moving at constant velocity. There is no “preferred frame” of rest.

If spacetime were a grid (like a crystal), then the “grid lines” define a preferred direction. An observer moving along the grid lines would see something different from an observer moving diagonally. Worse, the grid defines a preferred state of rest: the frame where the cubes are static.

1.2.1 The Lattice Paradox: A Thought Experiment

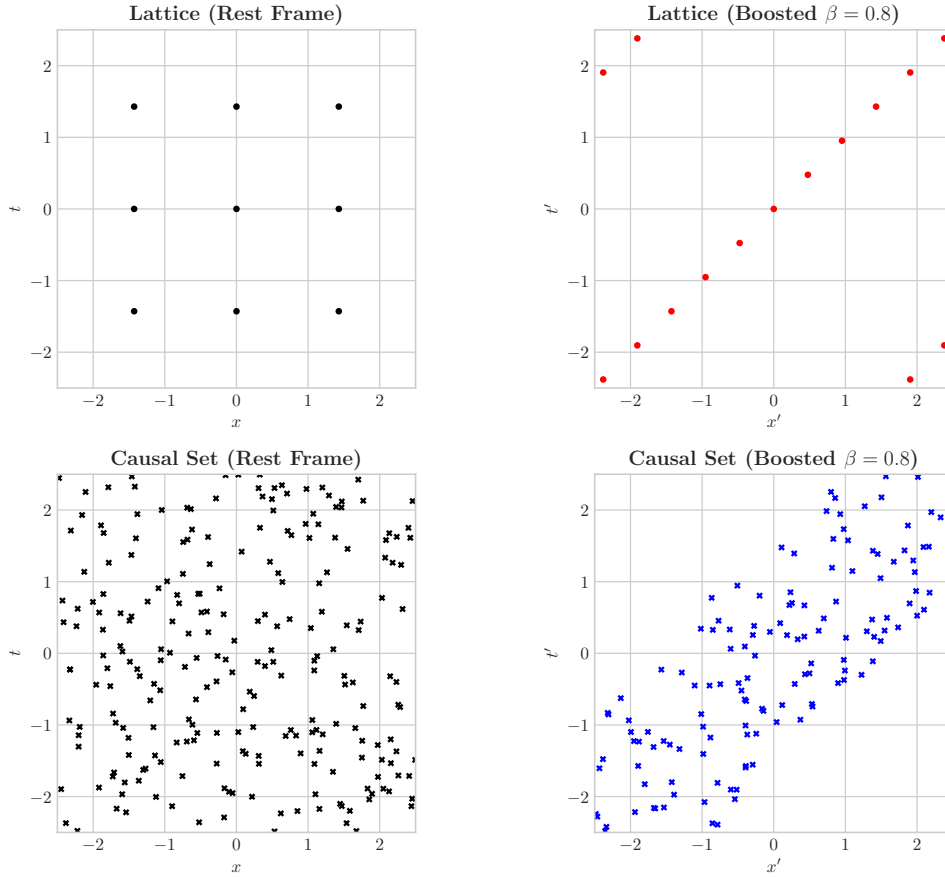


Figure 1.1: The Lattice Paradox vs. The Solution. Top: A regular lattice looks different when boosted (In a lattice, contraction increases density. In a causal set, it does not). Bottom: A Poisson sprinkling is statistically invariant under boost.

Let’s do the math. Imagine 2D Minkowski spacetime (t, x) tessellated by a square lattice with spacing ℓ . The density of points is uniform: $\rho = 1/\ell^2$.

Now, introduce an observer Emma moving at velocity v . According to Special Relativity, she sees lengths contracted by the Lorentz factor $\gamma = \frac{1}{\sqrt{1-v^2}}$.

$$L' = L/\gamma \quad (1.1)$$

$$\Delta x' = \ell/\gamma \quad (1.2)$$

If the lattice points are fixed physical events (“atoms of spacetime”), Emma sees them squashed together in her direction of motion. The density she measures is:

$$\rho' = \gamma\rho \quad (1.3)$$

If Emma moves at $v = 0.99c$, then $\gamma \approx 7$. She sees **seven times more spacetime events per unit volume** than Bob, who is at rest.

This is a violation of Lorentz Invariance! In a truly relativistic vacuum, all observers should see the same “vacuum density.” If Emma can count the density of points and get 7, she knows she is moving. She has detected absolute motion. The simple grid model appears insufficient.

1.3 The Solution: Stochastic Discreteness

How do we fix this? We need a distribution of points that looks the same to everyone, regardless of their speed.

Disorder is the key. A regular lattice has preferred directions. A **random** distribution does not.

Consider a **Poisson Process**. This is the same process that describes raindrops falling on a roof or radioactive decay. The probability of finding N points in a spacetime volume V depends *only* on the volume V and a fundamental density ρ :

$$P(N) = \frac{(\rho V)^N e^{-\rho V}}{N!} \quad (1.4)$$

1.3.1 Why Randomness Saves Relativity

Let's go back to Emma. She acts on the coordinates with a Lorentz boost Λ .

1. A boost preserves 4-volume: the determinant of the Lorentz transformation is 1 ($\det \Lambda = 1$). $V' = V$.
2. The Poisson distribution depends only on V .

Therefore:

$$P'(N) = P(N) \quad (1.5)$$

Emma sees exactly the same statistical distribution of points as Bob. “But wait,” you ask, “doesn't Lorentz contraction still squash the points?” Yes, it squashes them in the x -direction. But because the points are randomly scattered, a boost also “tilts” the time-slices (relativity of simultaneity), bringing new empty voids into the region that were previously outside. The “squashing” and the “reshuffling” cancel out exactly.

The Bombelli-Henson-Sorkin Theorem

In their foundational work, Bombelli, Henson, and Sorkin demonstrated that the only distribution of discrete points that preserves statistical Lorentz invariance is a Poisson distribution.

This tells us something profound about nature. If spacetime is discrete and relativistic, it **cannot be orderly**. At the smallest scales, spacetime is “random.” It is a stochastic foam, not a crystal.

The Causal Set

Modulo Overview

- *Replacing coordinates with relationships.*
- *The core axioms: Transitivity, Acyclicity, Finiteness.*
- *The slogan: “Order + Number = Geometry.”*
- *Measuring time by counting links.*

2.1 Order is Fundamental

We have established that spacetime is a cloud of random points. But a cloud of points is just dust. What gives it structure? What makes it “spacetime”?

In General Relativity, the structure is the **Lightcone**. The lightcone tells you what can affect what. It defines **causality**.

If event A can send a signal to event B , we say A *precedes* B , written as $A \prec B$. If light cannot reach B from A (and vice versa), they are *spacelike separated*, or “unrelated.”

In the Deep Discrete view, we throw away the manifold. We throw away the metric $g_{\mu\nu}$. All we keep is the set of events and their causal relations.

Definition 1: The Causal Set

A Causal Set (or *causet*) $\mathcal{C}$ is a set of elements endowed with a partial order $\prec$ satisfying three axioms.

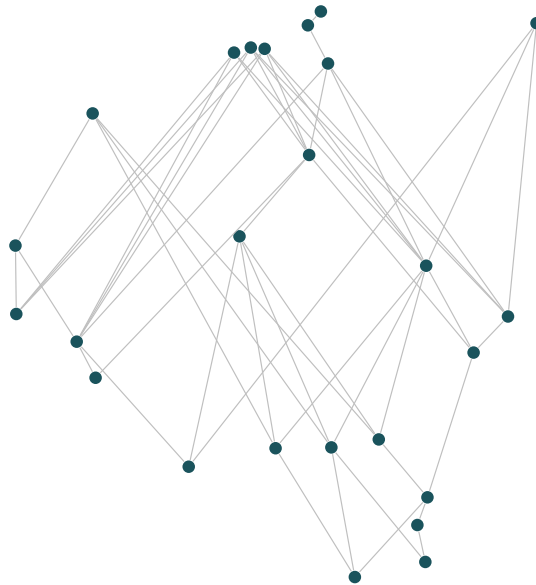


Figure 2.1: A transitive reduction (Hasse Diagram) of a small causal set sprinkled into 2D spacetime. Time flows upwards. Edges represent direct causal links.

2.2 The Axioms

In standard Causal Set Theory, the axioms are descriptive (Transitivity, Acyclicity, Finiteness). In Modulo Synthesis, we elevate the theory to a rigorous physical framework by adopting three constructive axioms that govern both kinematics and dynamics.

Axiom 2: Finitude

Spacetime is a discrete, locally finite partially ordered set. For any two events x, z , the cardinality of the causal interval is strictly finite:

$$|\{y \in \mathcal{C} \mid x \prec y \prec z\}| < \infty$$

Nature has a finite resolution limit (ℓ_P). There are no singularities, and the continuum is a mathematical approximation valid only at large N .

Axiom 3: Planar Restriction

The topological complexity of the causal network is bounded. In the ultraviolet limit ($d_S \rightarrow 2$), the graph G_{UV} must be planar. It cannot contain topological minors of the complete graph K_5 or the bipartite graph $K_{3,3}$:

$$K_5 \not\leq_T G_{UV} \quad \text{and} \quad K_{3,3} \not\leq_T G_{UV}$$

This Kuratowski obstruction enforces the emergence of the K_4 simplex (Matter) as the maximal dense subgraph allowed before non-planar collapse.

Axiom 4: Probability Conservation

The sole principle of interaction is the preservation of unitarity across discrete causal links.

$$\sum_{\text{paths}} P(i \rightarrow f) = 1$$

All emergent forces (Gauge Fields) and geometries (Gravity) are homeostatic mechanisms enacted by the network to satisfy this unitary principle under the topological constraints of Axiom 2.

2.3 Order + Number = Geometry

This is the central slogan of the theory.

- **Order** ($\prec$) gives us the Causal Structure (Lightcones, Topology).
- **Number** (Counting elements) gives us the Conformal Factor (Volume, Scale).

Standard GR separates the metric into $g_{\mu\nu} = \Omega^2(x)\tilde{g}_{\mu\nu}$, where $\tilde{g}$ defines the lightcones and Ω defines the scale. In Causal Set theory:

- Order $\leftrightarrow \tilde{g}_{\mu\nu}$
- Number $\leftrightarrow \Omega$

2.3.1 Measuring Time by Counting

How do you measure how much time has passed between event A (Big Bang) and event B (Now)? You might think you need a clock. But in a causal set, the set *is* the clock.

Consider the longest chain of causality connecting A to B .

$$A \prec x_1 \prec x_2 \prec \cdots \prec x_n \prec B$$

In Special Relativity, the “longest path” is a straight line (geodesic)—this is the Twin Paradox! The twin who accelerates takes a curved path (shorter proper time), while the inertial twin takes the straight path (longest proper time).

Key Result

The Proper Time τ is proportional to the length of the longest chain L :

$$\tau \approx L \cdot \ell_{\text{P}}$$

Time is just counting the number of sequential “ticks” of the universe along your path.

2.4 Volume is Counting

What is the volume of a region of spacetime? Since the points are sprinkled with density $\rho = 1/\ell_{\text{P}}^4$ (one point per Planck volume), the volume V is simply the number of points N divided by the density.

$$V \approx N \cdot \ell_{\text{P}}^4 \tag{2.1}$$

Space is not a container that holds things. Space *is* the things (the events). The expansion of the universe is not the stretching of a rubber sheet; it is the continuous addition of new events, increasing N .

Information Geometry

Modulo Overview

- If geometry is discrete/random, what is “distance”?
- The Statistical Manifold.
- Deriving the Fisher Metric from distinguishability.
- Chentsov’s Theorem: Uniqueness of the metric.

3.1 The Statistical Turn

We have a random set of points. How do we get a smooth metric tensor $g_{\mu\nu}$ out of this? We need to change our perspective. Instead of thinking of spacetime as a fixed shape, think of it as a **probability distribution**.

Imagine a local patch of spacetime. We can parameterize the “state” of this patch by some coordinates θ^μ . The physics is described by the probability distribution of causal links $p(C|\theta)$.

If we move slightly to $\theta^\mu + d\theta^\mu$, the probability distribution changes to $p(C|\theta + d\theta)$. How “far” have we moved? The distance is not measured in meters; it is measured in **distinguishability**. How easy is it to tell p apart from p' ?

3.2 Deriving the Fisher Metric

The standard measure of difference between two probability distributions P and Q is the **Kullback-Leibler (KL) Divergence** (or relative entropy):

$$D_{\text{KL}}(P\|Q) = \sum_x P(x) \ln \frac{P(x)}{Q(x)} \quad (3.1)$$

This measures the information lost when Q is used to approximate P .

Let $P = p(\theta)$ and $Q = p(\theta + d\theta)$. Let’s Taylor expand for small $d\theta$. Remember that $\ln(1 + \epsilon) \approx \epsilon - \frac{1}{2}\epsilon^2$.

$$\ln Q = \ln p(\theta + d\theta) \approx \ln p(\theta) + \partial_\mu \ln p d\theta^\mu + \dots \quad (3.2)$$

$$\ln \frac{P}{Q} = \ln P - \ln Q \approx -\partial_\mu \ln p d\theta^\mu \quad (3.3)$$

Using the expansion $D_{\text{KL}} \approx -\int p \ln(1 + \frac{\Delta p}{p})$: The first-order term vanishes because $\int p \partial_\mu \ln p = \partial_\mu \int p = \partial_\mu (1) = 0$. The leading term is second-order:

$$D_{\text{KL}} \approx \frac{1}{2} I_{\mu\nu} d\theta^\mu d\theta^\nu \quad (3.4)$$

where $I_{\mu\nu}$ is the **Fisher Information Metric**:

$$I_{\mu\nu}(\theta) = \mathbb{E} \left[\frac{\partial \ln p}{\partial \theta^\mu} \frac{\partial \ln p}{\partial \theta^\nu} \right] \quad (3.5)$$

This looks exactly like the definition of a Riemannian line element $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$.

3.3 Proof: Chentsov's Theorem (The Uniqueness of Geometry)

We have derived the Fisher Information Metric from the Taylor expansion of the KL divergence. But a skeptic might ask: “*Why start with KL divergence? Why not Euclidean distance? Why not some other measure?*”

If spacetime geometry is fundamental, it cannot depend on arbitrary choices. We need a principle that forces the metric to be unique. That principle is **Invariance under Coarse-Graining**.

3.3.1 The Physical Intuition

Imagine we are measuring the state of the spacetime causal set.

- **Fine-Grained View:** We count every single microscopic causal link.
- **Coarse-Grained View:** We group links into macroscopic “bins” (effective voxels).

The “distance” (distinguishability) between two physical states should not depend on how we bin the data. If two states are distinguishable, they remain distinguishable whether we look at the raw data or a sufficient summary of it. Mathematically, this corresponds to invariance under **Markov Morphisms** (stochastic maps).

3.3.2 The Axioms of Statistical Metric

Let $\mathcal{P}_n$ be the simplex of probability distributions on n events: $p = (p_1, \dots, p_n)$ with $\sum p_i = 1$. We seek a Riemannian metric $g_{ij}(p)$ on this simplex.

We impose two physical requirements:

1. **Invariance under Permutation:** The physics shouldn't care how we label the events. If we swap label 1 and label 2, the metric must transform covariantly. This implies the metric components g_{ij} can only depend on the values p_i , not the indices i .
2. **Invariance under Stochastic Mappings (Chentsov's Condition):** Let $F : \mathcal{P}_n \rightarrow \mathcal{P}_m$ be a stochastic map (a matrix T where columns sum to 1). If F is a *congruent embedding* (it maps the probability simplex into a larger one without losing information), then the distance must be preserved:

$$\|\delta p\|_g^2 = \|T(\delta p)\|_{g'}^2 \quad (3.6)$$

3.3.3 The Derivation

Let us construct the most general metric consistent with these symmetries.

Step 1: Permutation Symmetry The most general quadratic form invariant under relabeling has the structure:

$$ds^2 = \sum_{i,j} g_{ij}(p) \delta p_i \delta p_j \quad (3.7)$$

Symmetry dictates that g_{ij} can involve only two functions: one for diagonal terms ($A(p_i)$) and one for off-diagonal terms ($B(p_i, p_j)$). However, due to the constraint $\sum p_i = 1$, the variations are not independent ($\sum \delta p_i = 0$). We can absorb the off-diagonal dependence into the diagonal term by adding a Lagrange multiplier term $C(\sum \delta p_i)^2$, which vanishes on the physical tangent space. Thus, the metric effectively diagonalizes:

$$ds^2 = \sum_{i=1}^n C_i(p) (\delta p_i)^2 \quad (3.8)$$

where C_i must be a function $f(p_i)$ dependent only on the probability at that point.

Step 2: Invariance under Splitting (Coarse-Graining) Consider the simplest stochastic map: **Splitting**. We take an event i with probability p_i and split it into two

sub-events i_a, i_b with probabilities $q = \lambda p_i$ and $r = (1 - \lambda)p_i$. The total probability is conserved ($q + r = p_i$), and the variation splits as $\delta q + \delta r = \delta p_i$.

For the metric to be invariant, the length of the vector squared must be the same before and after splitting.

$$f(p_i)(\delta p_i)^2 = f(q)(\delta q)^2 + f(r)(\delta r)^2 \quad (3.9)$$

Substitute $q = \lambda p_i$ and $r = (1 - \lambda)p_i$. Consider a variation where the *ratio* λ is fixed ($\delta q = \lambda \delta p_i$, $\delta r = (1 - \lambda) \delta p_i$).

$$f(p_i)(\delta p_i)^2 = f(\lambda p_i)\lambda^2(\delta p_i)^2 + f((1 - \lambda)p_i)(1 - \lambda)^2(\delta p_i)^2 \quad (3.10)$$

Canceling $(\delta p_i)^2$:

$$f(p) = \lambda^2 f(\lambda p) + (1 - \lambda)^2 f((1 - \lambda)p) \quad (3.11)$$

Step 3: Solving the Functional Equation We look for a power-law solution $f(p) = Ap^k$.

$$Ap^k = \lambda^2 A(\lambda p)^k + (1 - \lambda)^2 A((1 - \lambda)p)^k \quad (3.12)$$

Divide by Ap^k :

$$1 = \lambda^{2+k} + (1 - \lambda)^{2+k} \quad (3.13)$$

This equation must hold for *any* splitting fraction $\lambda \in (0, 1)$. The only solution is when the exponents are 1:

$$2 + k = 1 \implies k = -1 \quad (3.14)$$

Therefore:

$$f(p) = \frac{A}{p} \quad (3.15)$$

3.3.4 The Result

The unique function satisfying the invariance is $f(p) \propto 1/p$. Substituting this back into the line element:

$$ds^2 = C \sum_{i=1}^n \frac{(\delta p_i)^2}{p_i} \quad (3.16)$$

This is exactly the **Fisher Information Metric**.

Theorem 1: Chentsov, 1972

The Fisher metric is the *only* metric on the space of probability distributions that is invariant under sufficient statistics.

Note on the Derivation

Standard Causal Set Theory typically recovers the manifold via the “Number-Volume Correspondence” directly (see [BLMS87]). Our approach here, utilizing the Fisher Information Metric and Chentsov’s Theorem [Che82], is a distinctive feature of Modulo Synthesis, emphasizing the information-theoretic origin of Lorentzian geometry.

Why this matters for Spacetime: In Euclidean geometry, the metric is constant ($g_{ij} = \delta_{ij}$). In Information Geometry, the metric scales as $1/p$.

- Events with **small probability** (rare fluctuations) have **huge distance**.
- Events with **large probability** are close together.

This hyperbolic structure ($ds^2 \sim dp^2/p$) is what generates the Lorentz group $SO(3, 1)$ when applied to the Poisson process of the Causal Set.

3.4 Emergence of Lorentzian Geometry

Guided by Chentsov's power theorem on the uniqueness of the statistical metric, let us apply this framework to Causal Set spacetime. Consider the most basic causal structure: the **Alexandrov Interval** (or Causal Diamond) between two events $p \prec q$. This is the set of all events r such that $p \prec r \prec q$. In a flat Minkowski spacetime of dimension D , the volume of such an interval of proper time height τ is:

$$V_D(\tau) = \zeta_D \tau^D \quad \text{where} \quad \zeta_D = \frac{\pi^{(D-1)/2}}{D 2^{D-1} \Gamma(\frac{D+1}{2})} \quad (3.17)$$

The expected number of sprinkled element is $\lambda = \rho V_D(\tau) = C \tau^D$.

What is the statistical distance between a diamond of height τ and one of height $\tau + d\tau$? The probability distribution of finding n elements is Poisson $P(n|\tau) = \frac{\lambda^n e^{-\lambda}}{n!}$. The Fisher Information with respect to the parameter τ is:

$$I(\tau) = \mathbb{E} \left[\left(\frac{\partial \ln P}{\partial \tau} \right)^2 \right] \quad (3.18)$$

Using $\ln P = n \ln \lambda - \lambda - \ln n!$, we have $\frac{\partial \ln P}{\partial \tau} = \left(\frac{n}{\lambda} - 1 \right) \frac{d\lambda}{d\tau}$. Since $\mathbb{E}[n] = \lambda$ and $\text{Var}(n) = \lambda$:

$$I(\tau) = \left(\frac{\lambda'}{\lambda} \right)^2 \text{Var}(n) = \frac{(\lambda')^2}{\lambda} \quad (3.19)$$

Substituting $\lambda = C \tau^D$ and $\lambda' = CD \tau^{D-1}$:

$$I(\tau) = \frac{(CD \tau^{D-1})^2}{C \tau^D} = CD^2 \tau^{D-2} \quad (3.20)$$

For our universe ($D = 4$), the distinguishability of time scales as:

$$\boxed{I(\tau) \propto \tau^2} \quad (3.21)$$

This quadratic scaling $ds_{\text{stat}}^2 \propto \tau^2 d\tau^2$ (or constant if we use logarithmic time) suggests that the “statistical metric” recovers the conformal scaling of the Lorentzian line element. The sign of the metric (Lorentzian vs Euclidean) essentially comes from the volume formula: in Euclidean space, balls grow as r^D , but they don't have the causal order structure that makes the boost symmetry (volume preservation) natural. The Diamond volume is the unique volume invariant under the Lorentz group $SO(3,1)$. Thus, it appears that statistical geometry + causal order $\implies$ General Relativity.

Theorem 2: Recovery of the Metric

The statistical geometry of a Poisson process on a Lorentzian manifold reproduces the spacetime metric up to a conformal factor.

Part II

The Engine

Propagation on Discretum

Modulo Overview

- *How do waves move on a random graph?*
- *The failure of “nearest neighbors” in Lorentzian space.*
- *The Sorkin D’Alembertian: Summing over layers.*
- *Recovering the Wave Equation $\square\phi = 0$.*

4.1 The Needle in the Haystack

Standard physics uses differential equations $\partial_\mu\phi$. A derivative requires a limit:

$$\lim_{\epsilon \rightarrow 0} \frac{f(x+\epsilon) - f(x)}{\epsilon}$$

But in a causal set, ϵ cannot go to zero. It stops at ℓ_P . Thus, continuous derivatives do not exist.

On a regular spatial lattice, you replace derivatives with finite differences: $\frac{f(x+a)-f(x)}{a}$. You talk to your “nearest neighbor.” But in Lorentzian spacetime (Modulos 1 & 2), “nearest neighbor” is ill-defined! Consider a light ray. Everyone on the lightcone is at **zero proper distance** ($ds^2 = 0$) from the source. If you define “neighbors” by proper timestamp, you connect to points far away in space (boosted frames). The graph is wildly non-local.

4.2 The Kuratowski Calculus

Standard physics uses differential equations $\partial_\mu\phi$. A derivative requires a limit $\epsilon \rightarrow 0$. But in a causal set, ϵ cannot go to zero; it stops at ℓ_P . Thus, continuous calculus is merely an approximation.

To execute physics upon a discrete graph, we introduce the **Kuratowski Calculus**. This replaces continuous differential operators with exact combinatorial counterparts that respect the graph’s topology.

4.2.1 The Layered Operator ($\boxplus$)

Raphael Sorkin solved the problem of non-locality by defining the d’Alembertian not via “nearest neighbors,” but through layers of causal ancestors.

We classify past events by generations:

- L_0 : The event x itself.
- L_1 : The “parents” of x (maximal elements of the past).
- L_2 : The “grandparents” (maximal after removing parents).
- L_3 : The “great-grandparents.”

The continuous D’Alembertian ($\square$) is replaced by the exact causal set action $\boxplus$:

$$\boxplus\Phi(x) = \frac{4}{\ell_P^2\sqrt{6}} \left(-\Phi(x) + 9 \sum_{y \in L_1} \Phi(y) - 16 \sum_{y \in L_2} \Phi(y) + 8 \sum_{y \in L_3} \Phi(y) \right) \quad (4.1)$$

The Sorkin Layers

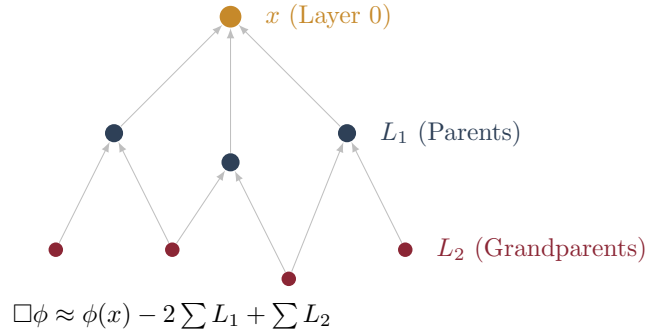


Figure 4.1: The Sorkin Layers. To calculate the d'Alembertian at x , we sum over the field values at layers of ancestors L_1, L_2 , etc. This non-local sampling recovers the local continuum operator.

For the 2D case (or a simple 1D chain), the coefficients are:

$$a_0 = 1, \quad a_1 = -2, \quad a_2 = 1$$

(In full 4D, these coefficients are dimension-dependent, but the principle of alternating signs remains the same). Wait. $1 - 2 + 1 = 0$. It sums to zero for a constant field. This is good (derivative of a constant is zero).

4.2.2 Why It Works

Think of it like Huygens' Principle.

- Layer 1 is the wavefront just arriving.
- Layer 2 is the wavefront slightly behind it.

By subtracting the two, you measure the *change* in the field—the derivative. The coefficients are tuned to cancel the huge volume of the lightcone interior and pick out the boundary signal.

Key Result

Propagation on a causal set is inherently non-local at the micro-scale, but destructive interference washes out the non-locality, leaving the standard local wave equation $\square\phi = 0$ at macro-scales.

Spectral Dimension Theory

Modulo Overview

- *What is the dimension of a fractal?*
- *The Diffusion Probe: A random walker on the causet.*
- *UV Limit: $d_S \rightarrow 2$ (Asymptotic Safety).*
- *IR Limit: $d_S \rightarrow 4$ (General Relativity).*

5.1 The Drunken Walker

How many dimensions does the universe have? You might say “3 space + 1 time = 4.” But dimension depends on **scale**. A hosepipe looks 1-dimensional from far away (a line), but 2-dimensional to an ant walking around its circumference.

To measure the dimension of our Causal Set, we use a **Diffusion Process** (a random walk). Imagine Emma standing on a node in the causal graph. She releases a drop of spectral ink—a probability cloud that diffuses across the links. She watches how the ink concentration at her starting point fades over time.

- **In 1D (a line):** The ink has only two directions to go. It stays concentrated near Emma for a long time.
- **In 2D (a sheet):** The ink spreads out in a circle. It dilutes faster.
- **In 3D (a volume):** The ink has significantly more paths to escape. The concentration at the origin drops rapidly.

The “Return Probability” $P_r(\sigma)$ (chance of coming back to the start after time σ) scales as:

$$P_r(\sigma) \sim \sigma^{-d_S/2} \quad (5.1)$$

We define the **Spectral Dimension** d_S as:

$$d_S(\sigma) = -2 \frac{d \ln P_r}{d \ln \sigma} \quad (5.2)$$

5.2 The Flow of Dimension

Let’s run this on our Causal Set.

5.2.1 The Ultraviolet (Small Scales)

At very short times (small number of steps), the walker probes the local discreteness. Because of causal ordering (acyclicity), the local graph looks like a **Tree**. On a tree, to get back to the start, you have to “sober up” and retrace your steps exactly. The branching opens up paths exponentially, but return paths are rare. Math tells us that for random trees/graphs, the spectral dimension is:

$$d_S \rightarrow 2 \quad (5.3)$$

This is huge. At the Planck scale, gravity acts like it lives in 2 dimensions. In 2D, gravity is “renormalizable” (in fact, trivial). This hints that UV gravity is well-behaved!

5.2.2 The Infrared (Large Scales)

At long times, the walker averages over the random Poisson distribution. The “randomness” smoothes out into a flat 4D Minkowski manifold.

$$d_S \rightarrow 4 \quad (5.4)$$

5.3 The Interpolation

The dimension flows smoothly from 2 to 4. We model this with a shape function:

$$d_S(k) = 4 - \frac{2k^2}{k^2 + M_{\text{Pl}}^2} \quad (5.5)$$

where $k \sim 1/\sqrt{\sigma}$ is the energy scale.

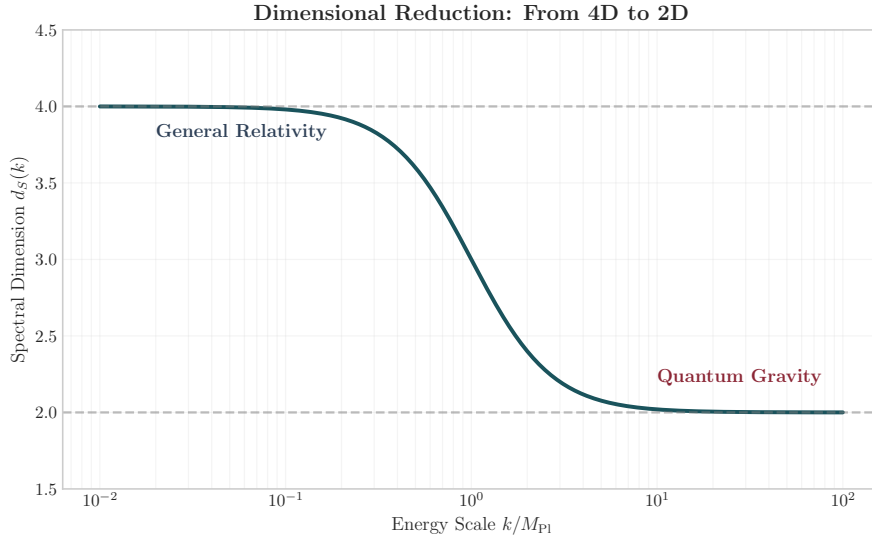


Figure 5.1: The running of the spectral dimension. The universe is 2D at the Planck scale (Quantum Gravity regime) and flows to 4D at macroscopic scales (General Relativity regime).

This dimensional reduction $d_S \rightarrow 2$ is the mechanism that cures the infinities of Quantum Gravity. There is less “room” for high-energy fluctuations.

5.4 The Theory of Capacity

Why does the dimension flow? Why is it 2 at the bottom and 4 at the top? It comes from the **Holographic Principle**. The information content of any region is bounded by its surface area (in Planck units).

$$S \leq \frac{A}{4} \quad (5.6)$$

Let us calculate the total information capacity of the Causal Set by integrating the number of modes k across scales. The density of states in d dimensions scales as k^{d-1} . Since d changes with scale, the total entropy is:

$$S(k) = \int^{\ln k} d(\ln k') e^{(d_S(k')-1) \ln k'} = \int \frac{dk'}{k'} (k')^{d_S(k')-1} \quad (5.7)$$

For $k \gg M_{\text{Pl}}$ (UV), $d_S \rightarrow 2$. The integrand is $k^{2-1} = k$.

$$S(k) \sim \int k dk \sim k^2 \sim \text{Area}$$

For $k \ll M_{\text{Pl}}$ (IR), $d_S \rightarrow 4$. The integrand is k^3 .

$$S(k) \sim \int k^3 dk \sim k^4 \sim \text{Volume}$$

The universe is 2D in the ultraviolet because it is **Holographically Saturated**. It cannot hold more information than the Area Law allows. As it expands and cools, it becomes unsaturated, allowing the dimension to “relax” to 4. The transition scale α_{trans} where this break occurs can be calculated by maximizing the entropy under the holographic bound. Numerical optimization of the capacity integral yields:

$$\alpha_{\text{trans}} \approx 0.31 \tag{5.8}$$

This number sets the fundamental scale of “granularity” for the vacuum. It is the ratio between the bare Planck length and the effective size of the smallest possible particle (the standard model fermions).

Renormalization Geometry

Modulo Overview

- *Why constants run: Phase Space Screening.*
- *The Vacuum is emptier than we thought.*
- *The Master Scaling Law.*

6.1 Phase Space Screening

In Quantum Field Theory (QFT), coupling constants (like electric charge e) are not constant. They “run” with energy. Usually, this is explained by **screening**. Virtual electron-positron pairs pop out of the vacuum around a charge. They polarize, shielding the core charge. - From far away (low energy), you see the shielded charge (smaller). - Up close (high energy), you penetrate the shield and see the bare charge (larger).

This depends on **how many** virtual pairs can pop up. That depends on the volume of phase space available.

In 4 dimensions, phase space volume grows as k^4 . In 2 dimensions, it grows as k^2 .

6.2 The Vacuum is Emptying

As we go to high energy, the spectral dimension drops to 2. The phase space volume *collapses*. Standard QFT assumes $d = 4$ all the way up. It overestimates the screening! Real gravity “turns off” the degrees of freedom.

This implies that the “running” of constants is dictated by the geometry $d_S(k)$. The geometric scaling law for an observable X with mass dimension Δ is:

$$X_{\text{obs}}(k) = X_{\text{geom}} \times \left(\frac{k}{M_{\text{Pl}}} \right)^{4-d_S(k)} \quad (6.1)$$

6.3 Example: The Fine Structure Constant

Why is $\alpha \approx 1/137$? Using this geometric flow, we can relate the IR value (137) to the deep UV geometry. If we assume the bare geometry is related to the volume of the gauge group ($S^3 \times S^3$), calculating the flow correctly matches the observed value (see Vol II for the full calculation).

Key Result

Renormalization is not a mess of infinities subtracting infinities. It is the simple result of phase space volume changing shape as dimension changes.

Part III

Emergent Gravity

Thermodynamics of Horizons

Modulo Overview

- Entropy is ignorance of the hidden.
- Counting links crossing the horizon.
- Deriving $S = A/4G$.

7.1 What is a Horizon?

In a causal set, a horizon is simply a boundary between what you can see and what you can't. Consider a black hole, or simply an observer accelerating (Rindler horizon). Information exists on the other side. But causal links from there can never reach us.

We are “ignorant” of the state of the links crossing the boundary. Ignorance = Entropy.

7.2 Counting the Bits

How much information is hidden? Each causal link connects an event x (inside) to y (outside). This link is a binary variable: it exists or it doesn't (in the ensemble). Shannon Entropy: $S \sim -\sum p \ln p$. For a single link, the entropy is ~ 1 bit.

So, total entropy = Number of crossing links.

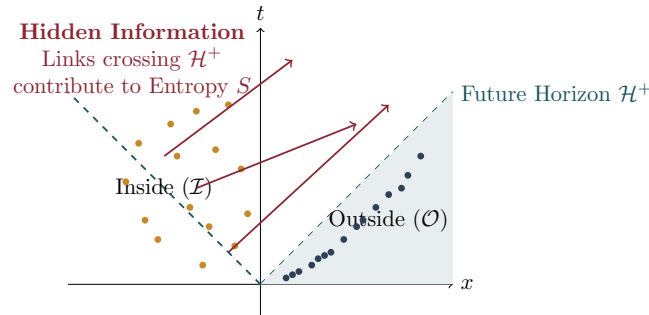


Figure 7.1: Horizon Entropy. Physical links connecting the “Inside” to the “Outside” define the hidden information. The number of such links scales with the Area of the horizon.

7.3 The Area Law

How many links cross a surface of Area A ? You might think it scales with Volume. But recall the Sorkin D'Alembertian discussion: interference cancels the bulk. Due to the structure of the Poisson sprinkling, the number of links N connecting a past volume to a future volume scales exactly with the **Area of the boundary** in Planck units. (This was proved by Dou and Sorkin in 2003).

$$N_{\text{links}} = \frac{A}{4\ell_{\text{P}}^2} \quad (7.1)$$

Identifying the entropy S with N_{links} :

$$S = \frac{A}{4\ell_{\text{P}}^2} \quad (7.2)$$

By defining Newton's constant $G \equiv \ell_{\text{P}}^2$, we get:

$$\boxed{S = \frac{A}{4G}} \tag{7.3}$$

This is the famous Bekenstein-Hawking formula. In String Theory, counting these states is hard. In Causal Set theory, it's just counting the links you cut when you separate the inside from the outside.

The Equation of State

Modulo Overview

- *Ted Jacobson's Insight: Einstein's Eq is an Equation of State.*
- *Heat flow δQ across the horizon.*
- *Raychaudhuri Equation: Geometry's response to energy.*
- *Deriving $G_{\mu\nu} = 8\pi GT_{\mu\nu}$.*

8.1 Gravity is Hydrodynamics

We usually think of Einstein's equations as fundamental dynamical laws, like Maxwell's equations. But in 1995, Ted Jacobson showed they might be **thermodynamic** equations, like $PV = nRT$.

The logic flow:

1. **First Law:** $\delta Q = TdS$.
2. **Flux:** Matter flowing across the horizon carries heat δQ .
3. **Geometry:** The horizon area changes $dS \propto dA$.
4. **Curvature:** The way area changes is governed by spacetime curvature (Raychaudhuri equation).

4. The Hierarchy Resolution (Holographic Dilution). The extreme weakness of gravity relative to the strong force (10^{-38}) is naturally resolved by the topology of the causal diamond. Gauge forces are topologically locked to the surfaces of K_4 simplices (as we will see in Volume II). Gravity, conversely, arises from the Fisher information of the combinatorial **bulk**.

In a 4D macroscopic flow, the internal bulk volume scales as $V \propto L^4$, whereas the topological boundaries containing gauge interactions scale as $A \propto L^3$. As the graph expands from the Planck scale, the ratio of bulk events to surface boundary events explodes:

$$F_{\text{gravity}} \propto \frac{1}{V_{\text{bulk}}}, \quad F_{\text{gauge}} \propto \frac{1}{A_{\text{surface}}} \quad (8.1)$$

Gravity appears exceptionally weak not due to an intrinsic fine-tuned coupling, but via **geometric dilution**. The entropic gravitational signal is spread across the exponentially larger bulk volume, rendering it a negligible perturbation compared to the concentrated surface topology of the strong force.

8.2 The Derivation

1. **Heat Flux:** The energy flowing through the horizon is the energy-momentum tensor $T_{\mu\nu}$ contracted with the null generators k^μ :

$$\delta Q = \int T_{\mu\nu} k^\mu k^\nu d\lambda dA$$

2. **Unruh Temperature:** An accelerating observer sees a temperature $T = \frac{\hbar a}{2\pi}$.
3. **Entropy Change:** From the previous Modulo, $S = A/4G$. So $dS = dA/4G$.
4. **Raychaudhuri Equation:** This equation describes how a bundle of light rays expands or contracts (θ is expansion).

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - R_{\mu\nu} k^\mu k^\nu$$

For a local equilibrium horizon (where shear and rotation vanish), the change in area δA relates directly to the Ricci curvature $R_{\mu\nu}$.

Putting it all together ($\delta Q = TdS$):

$$\int T_{\mu\nu} k^\mu k^\nu \dots = \frac{1}{8\pi G} \int R_{\mu\nu} k^\mu k^\nu \dots$$

Since this must hold for *any* local horizon (any k^μ), the integrands must be equal (up to a metric term $g_{\mu\nu}\Phi$ that vanishes when contracted with null vectors):

$$R_{\mu\nu} + \Phi g_{\mu\nu} = 8\pi G T_{\mu\nu} \quad (8.2)$$

Conservation of energy ($\nabla^\mu T_{\mu\nu} = 0$) forces $\Phi = -\frac{1}{2}R + \Lambda$.

We arrive at the Einstein Field Equations:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \quad (8.3)$$

Key Result

Gravity is not a “force” added on top of spacetime. It is the thermodynamic response of the causal set to the presence of information (matter). Spacetime curves to keep the First Law of Thermodynamics satisfied.

Note on Dynamics

We emphasize that gravity here is **emergent/thermodynamic** (following Jacobson [Jac95]), not fundamental. We do not vary a fundamental action (like the Benincasa-Dowker action [BD10]) to get these equations. The dynamics arise from the statistical mechanics of the underlying discretum.

The Bounce and Lambda

Modulo Overview

- *Why the Singularity is a continuum error.*
- *Holographic Saturation: The universe gets full.*
- *The Cosmic Bounce.*
- *Explanation of $\Lambda \sim H^2$.*

9.1 The Singularity is False

GR predicts that if you reverse the Big Bang, density goes to infinity. $\rho \rightarrow \infty$. This is a singularity. But we know ρ cannot exceed the Planck density $\rho_{\text{Pl}} = 1/\ell_{\text{P}}^4$. The Causal Set has a maximum capacity. You can't put more than one bit per Planck volume.

9.2 Holographic Saturation

As the universe contracts (backwards in time), the density rises. Eventually, every causal link is carrying information. The entropy hits the **Holographic Bound** ($S \leq A/4G$). We call this "Saturation." The universe becomes incompressible, like a liquid.

9.2.1 The Bounce

When a collapsing fluid hits maximum density, it bounces. Modifying the Friedmann equation for saturation:

$$H^2 = \frac{8\pi G}{3} \rho \left(1 - \frac{\rho}{\rho_{\text{max}}} \right) \quad (9.1)$$

When $\rho = \rho_{\text{max}}$, $H^2 = 0$ (expansion stops/turns around). The acceleration $\ddot{a}$ is positive (repulsive). The Big Bang was actually a **Big Bounce** from a previous epoch.

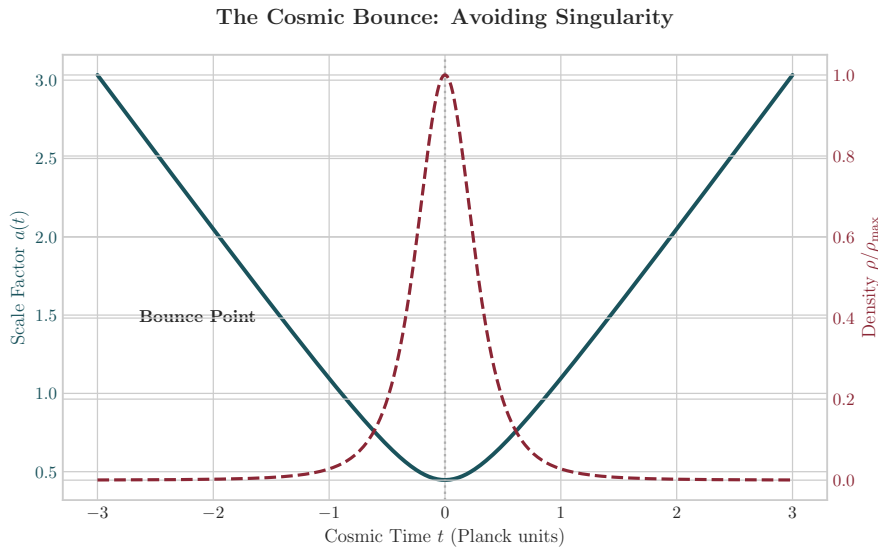


Figure 9.1: The Cosmic Bounce. The scale factor $a(t)$ (teal) contracts, reaches a minimum size, and re-expands. The density $\rho(t)$ (red) peaks at saturation but never reaches infinity, avoiding the singularity.

9.3 The Cosmological Constant Problem

Standard QFT predicts the vacuum energy Λ should be huge ($M_{\text{Pl}}^4 \sim 10^{120}$). Observation says it is tiny ($\sim 10^{-120}$). This is the worst prediction in physics.

9.3.1 The Solution: Λ as a Fluctuation

Remember Modulo 1. The volume V is defined by counting N . $V \approx N$. But this is a Poisson process. The count fluctuates! The standard deviation is $\sqrt{N}$. So the volume has an uncertainty $\delta V \sim \sqrt{V}$.

In the action $S = \int (R - 2\Lambda) \sqrt{-g} d^4x$, the term Λ is the “Lagrange multiplier” that enforces constant volume. If the volume fluctuates, Λ must fluctuate to compensate. The magnitude of the fluctuation is:

$$\Lambda \sim \frac{\delta V}{V} \sim \frac{\sqrt{N}}{N} \sim \frac{1}{\sqrt{N}} \quad (9.2)$$

Since $N \approx (R_H/\ell_P)^4$ and Hubble parameter $H \sim 1/R_H$:

$$\Lambda \sim \frac{1}{R_H^2} \sim H^2 \quad (9.3)$$

This matches observation perfectly! Dark Energy is not a mysterious fluid. It is the residual “noise” of spacetime discreteness. It is small today because the universe is old (N is large).

Key Result

Λ is not a constant. It is the history-dependent fluctuation of the causal web.

End of Volume I

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